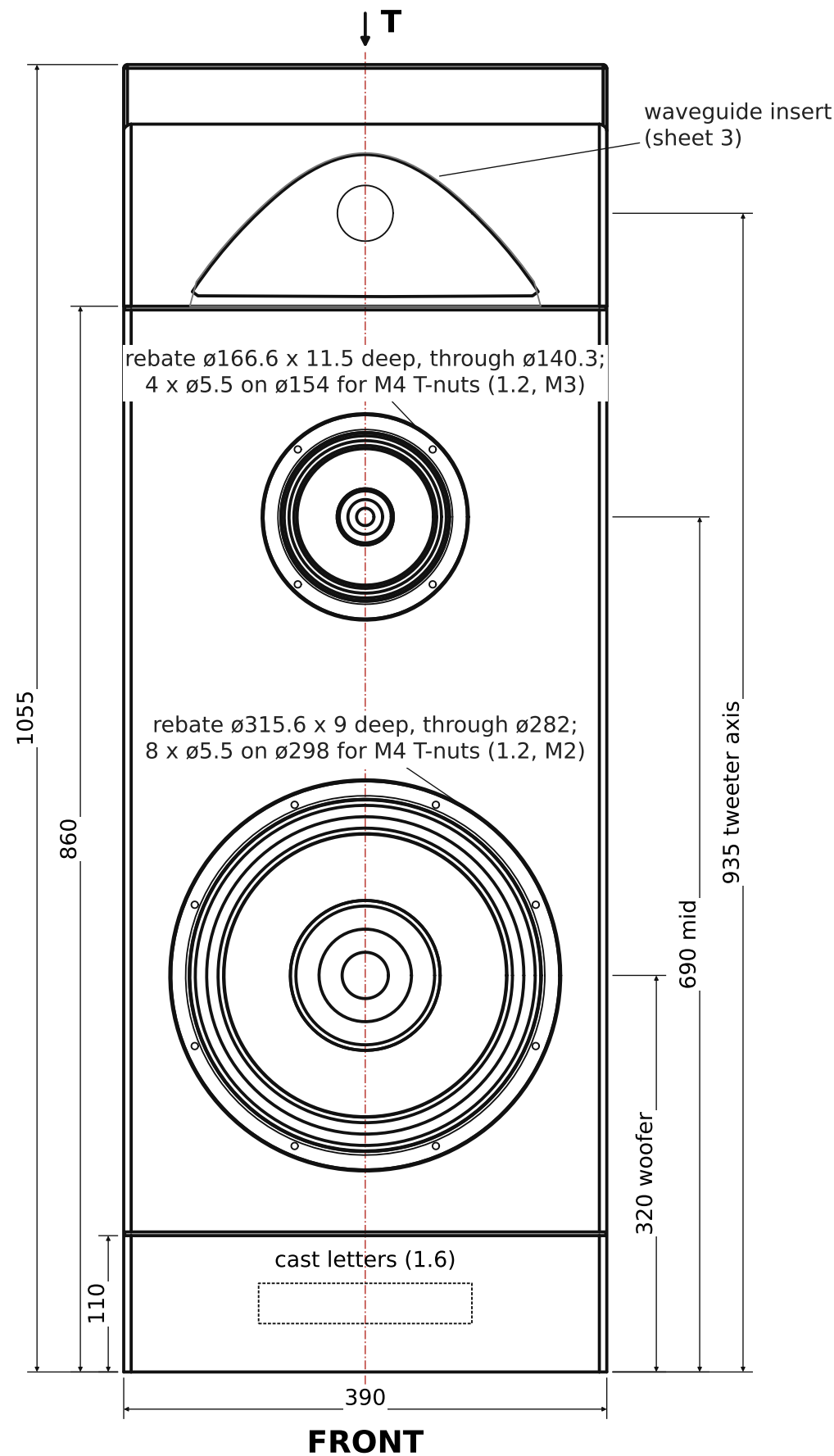
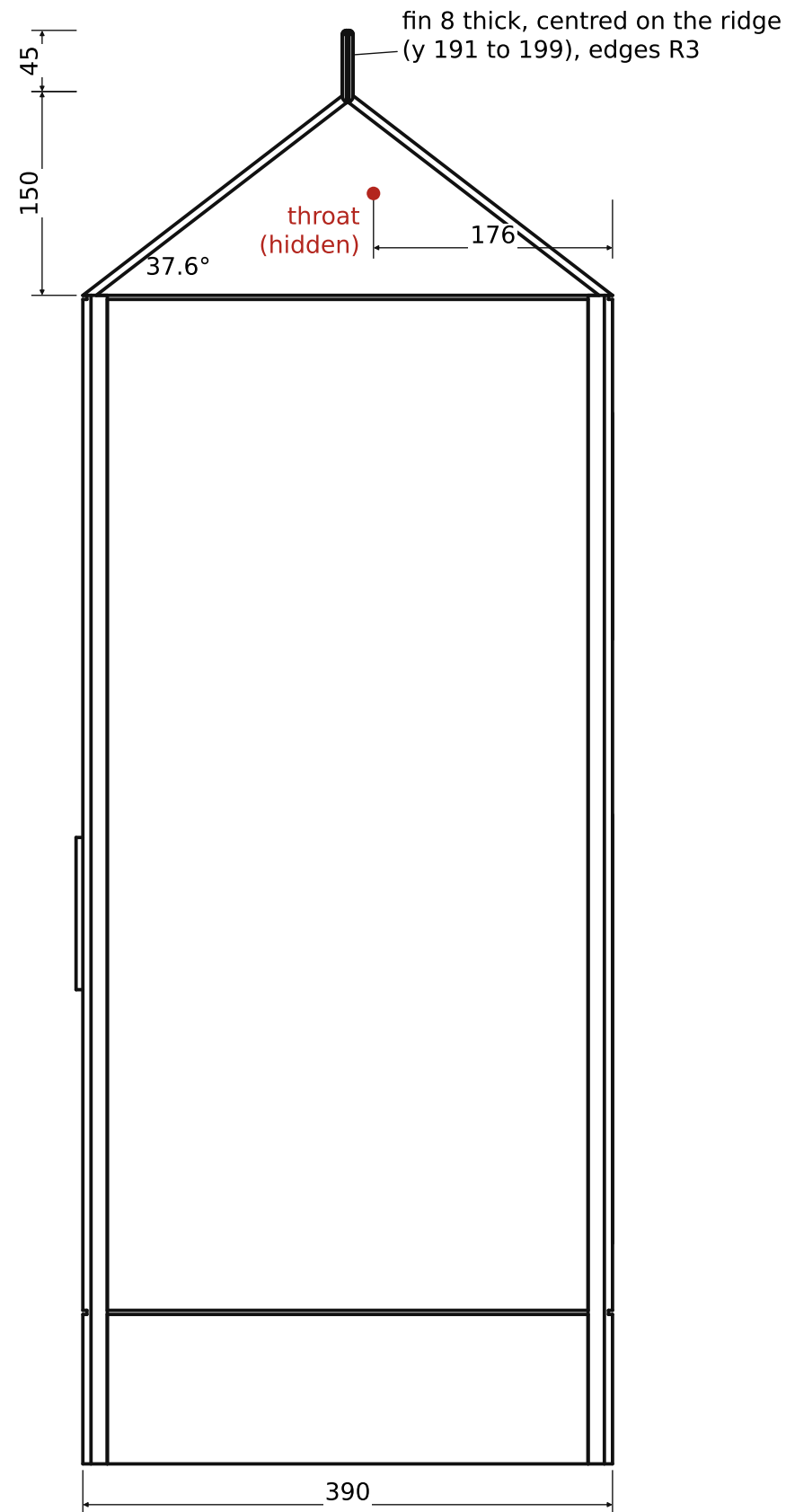
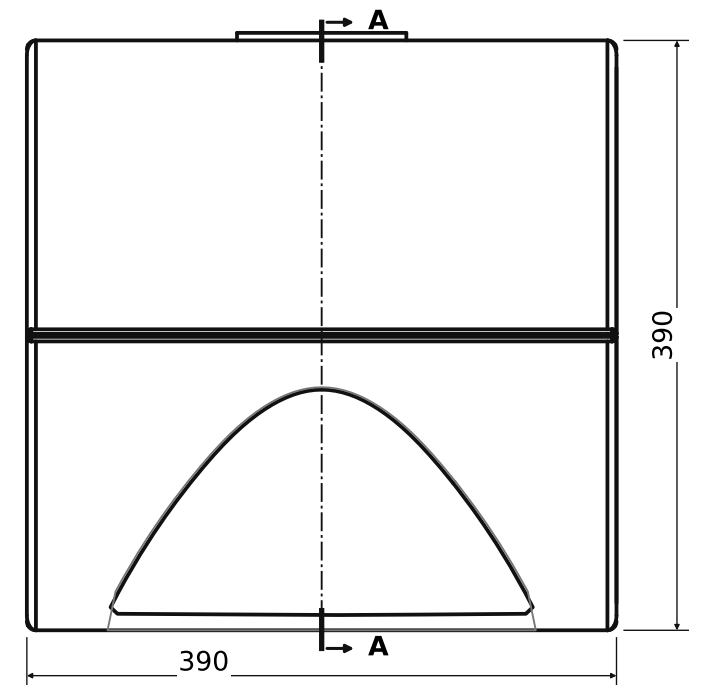




FRONT



LEFT SIDE



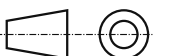
VIEW T (FROM ABOVE)

black: the waveguide's mouth, where its lip meets the roof
grey: the insert's seam

Notes: notes 1.1 to 1.6, sheet 7

General arrangement

earmilk floorstander. Front, left side and top at 1:5. Heights from the floor; depths from the front face.

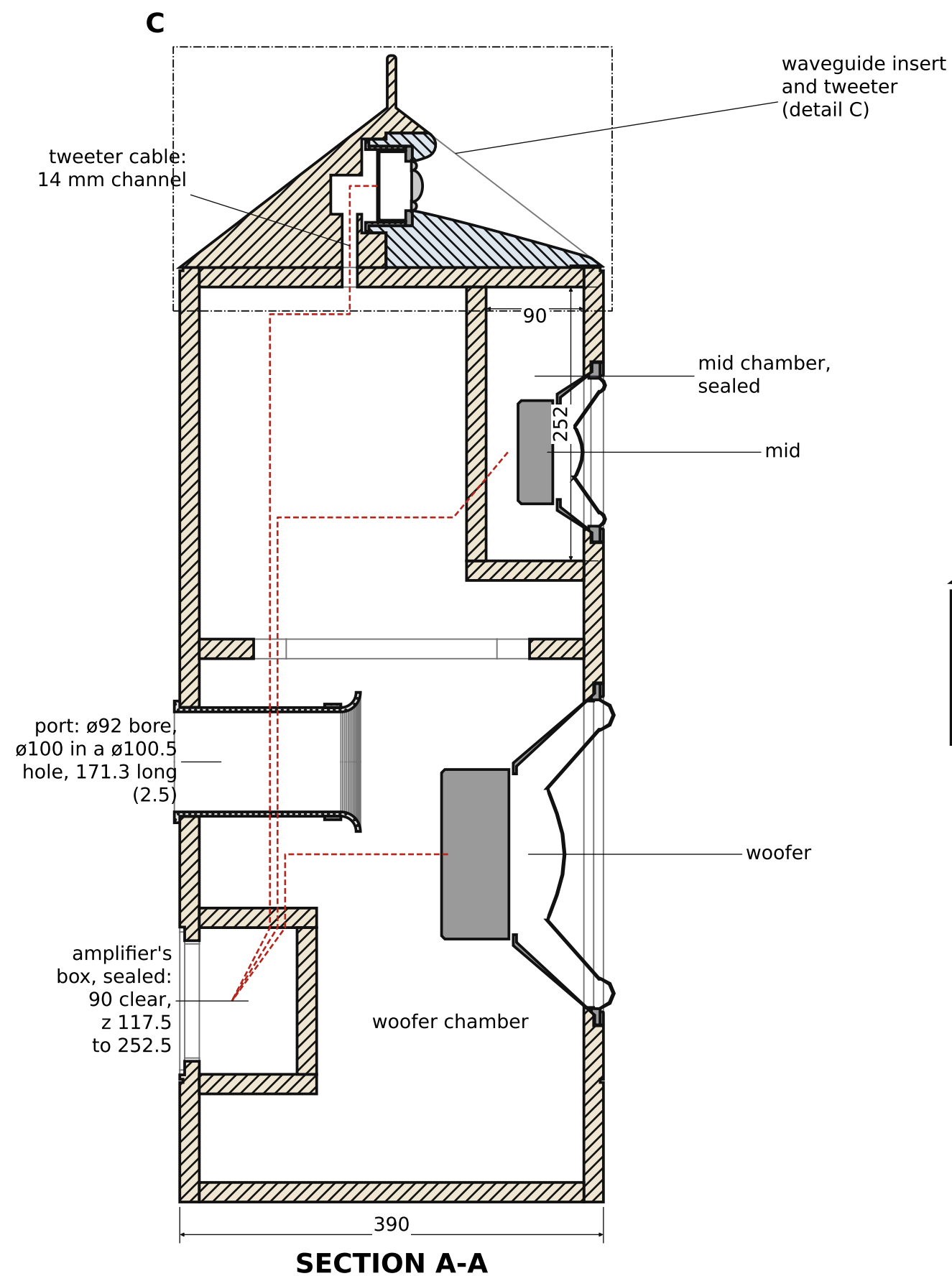


mm | issue 2026-10-09 | from fab/sheets.py (the CAD)

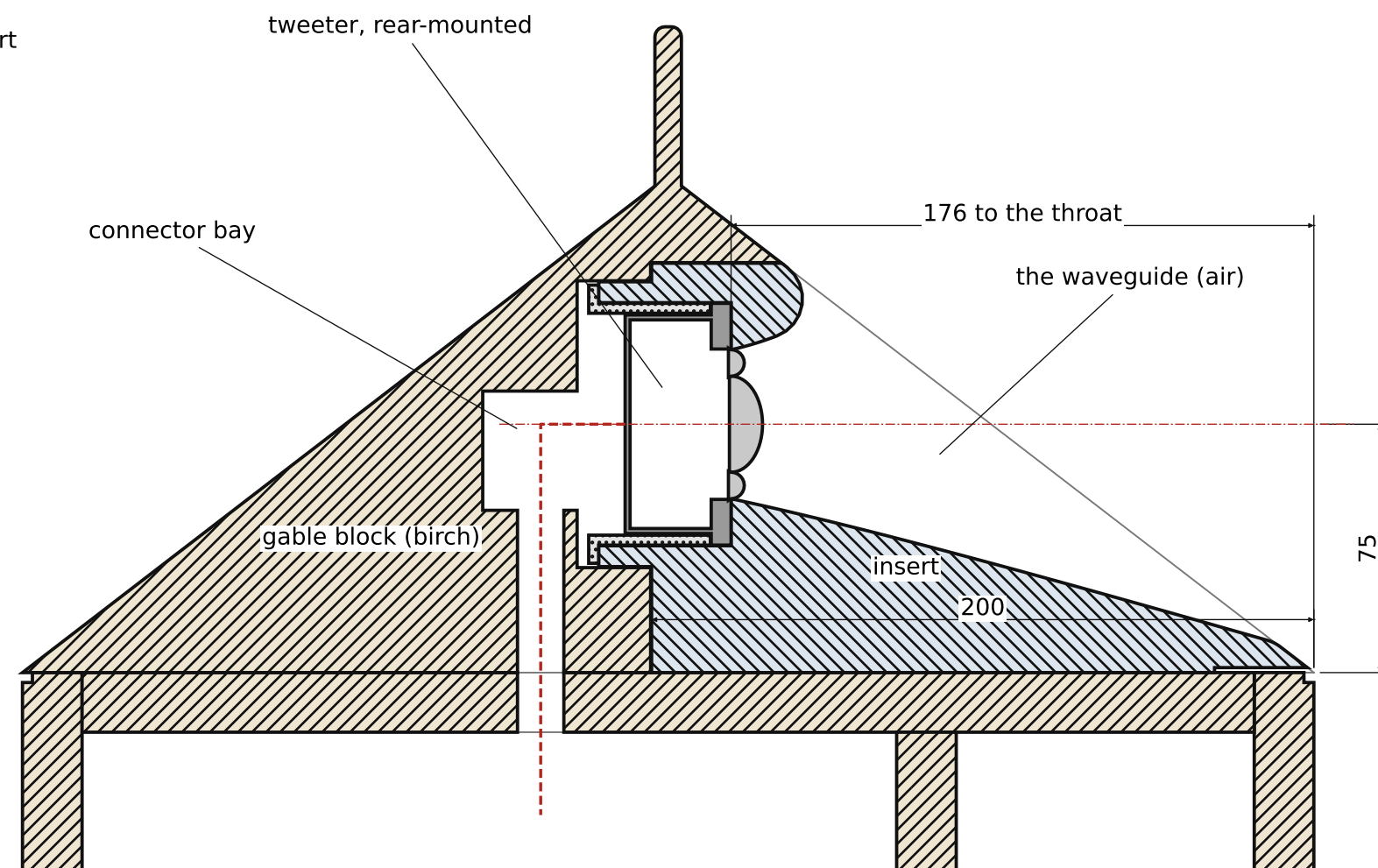
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

1 of 8

EM-FS-001 rev D



red dashed: the cables' runs to the amplifier (sheet 4)



The roof: notes 2.1 to 2.5, sheet 7

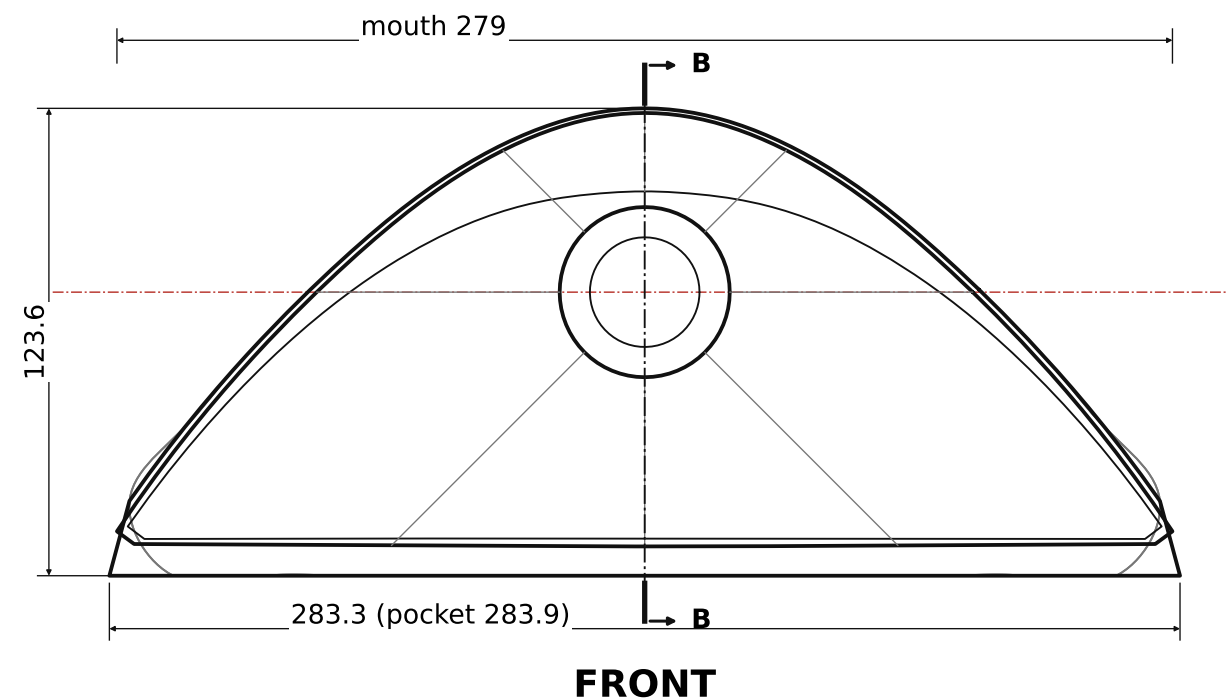
Section on the centreline

earmilk floorstander. Section A-A on the centre plane (x 195), from the left, at 1:5; the roof at 1:2. The insert hatched the other way.

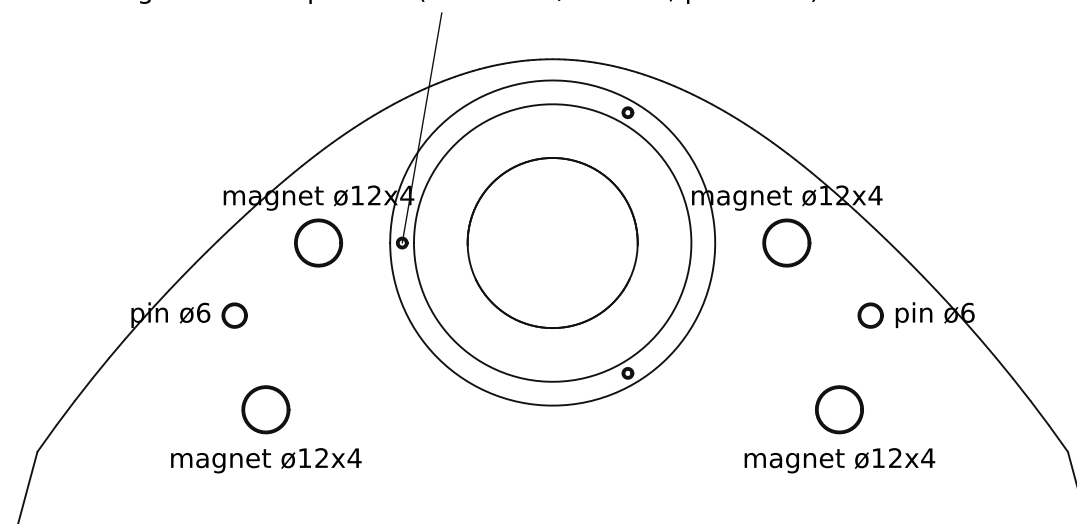


mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

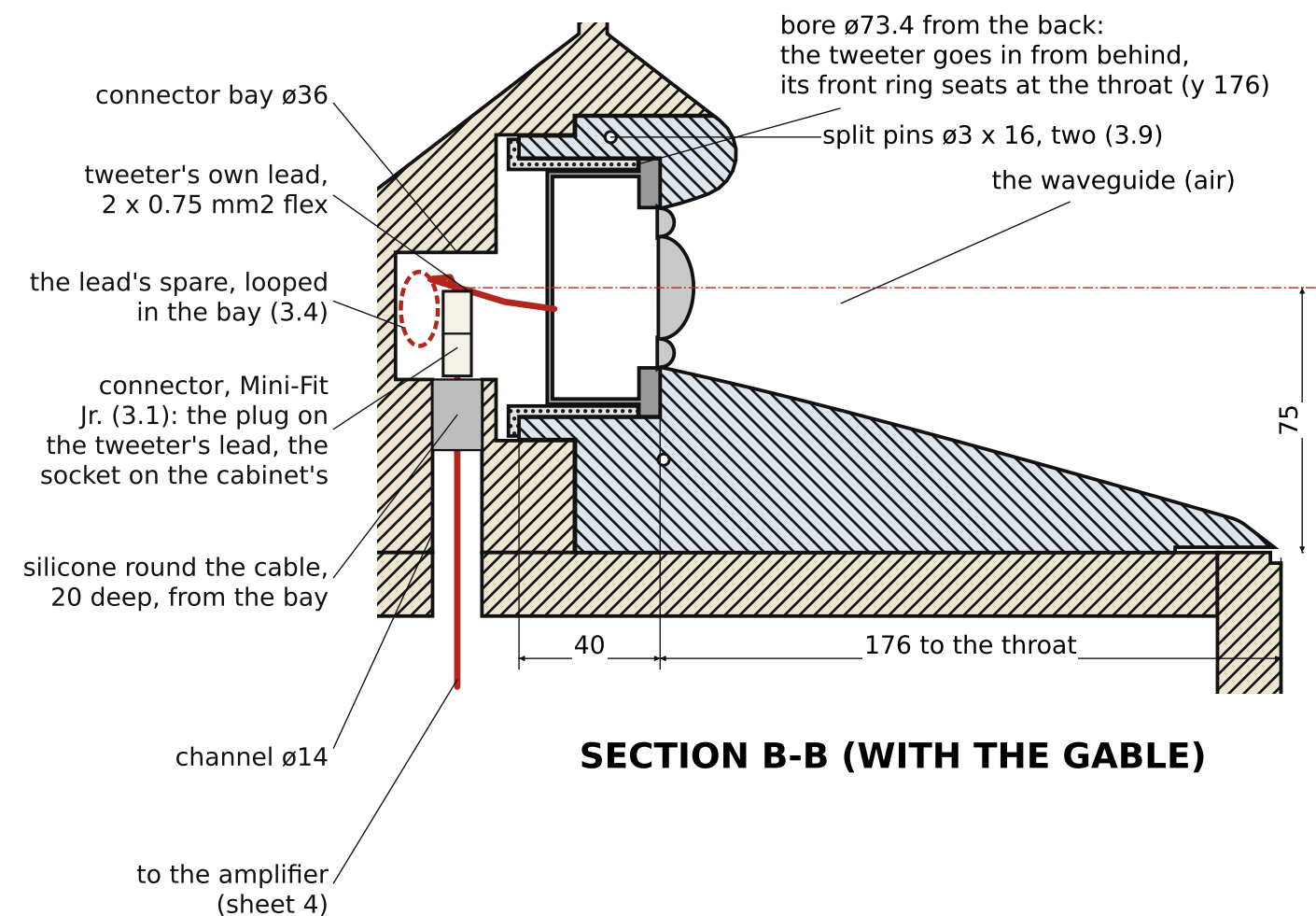
2 of 8
EM-FS-002 rev D



3 x $\varnothing 2.4 \times 10$ pilots on $\varnothing 79.6$ in the boss's back face, one on the horizontal at the speaker's right (left here), 120° apart, for the retaining sleeve's 3.0 x 12 thread-forming screws for plastics (WN 1411 / PT K30, pan head)



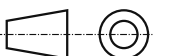
BACK (THE FACE THAT MEETS THE POCKET)



Fitting the tweeter and the insert: notes 3.1 to 3.9, sheet 7

Waveguide insert and tweeter mount

earmilk floorstander. The insert at 1:2: front, section on the axis, back; the pocket; fitting and service.

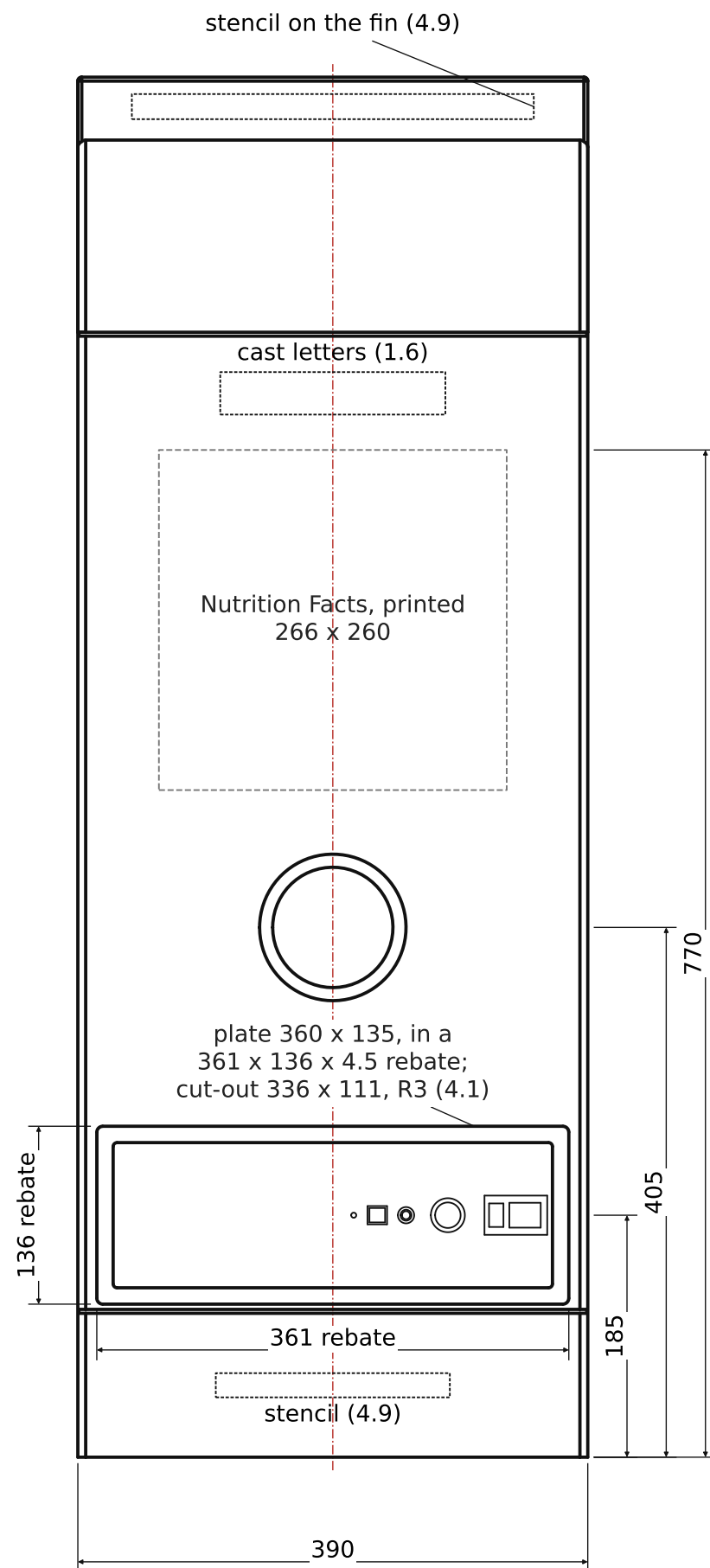


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Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

3 of 8

EM-FS-003 rev D

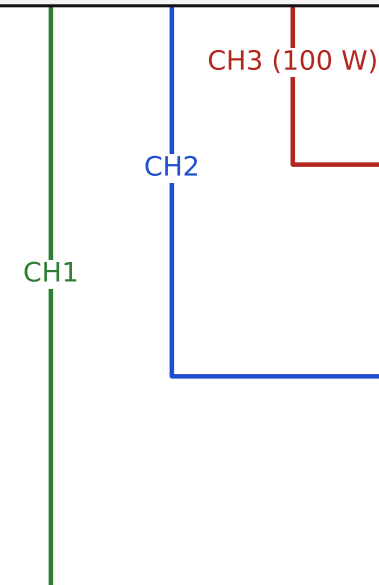


BACK

Mains
IEC C14 inlet, switch

Signal in
XLR or RCA
USB for the DSP

Hypex FusionAmp FA253
3 channels, DSP crossover and EQ (ADAU1452)
250 + 250 W into 4 ohm, 100 W tweeter channel
on its side across the back's foot
in its own sealed box (not airtight itself)
filters: Hypex Filter Design, over USB
mains in on the plate (IEC C14 with switch)



Tweeter, SB Satori TW29DN-B
29 mm dome, faceplate off, on the insert
1 mm2 pair, red +, black -

Midrange, SB Satori MR16P-8
6.5 in papyrus, its own sealed chamber
1.5 mm2 pair, red +, black -

Woofer, Dayton Audio RSS315HF-4
12 in, 4 ohm, vented to 32 Hz
2.5 mm2 pair, red +, black -

Runs, connectors and seals: notes 4.1 to 4.9, sheet 8

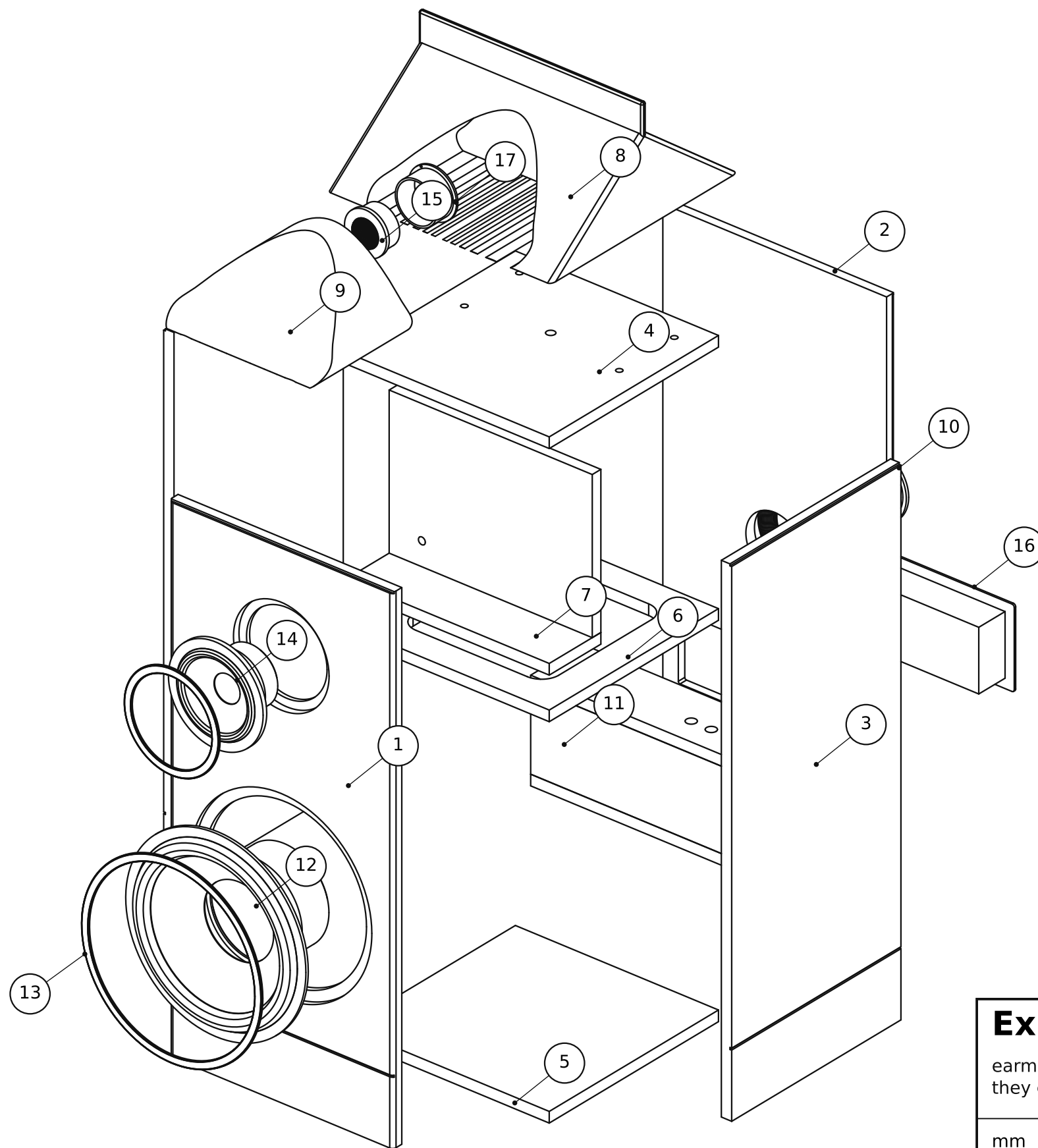
Back and wiring

earmilk floorstander. The back at 1:5 and the amplifier, drivers, cables and connectors. Active: the amplifier's DSP is the crossover.



mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

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EM-FS-004 rev D



Parts

- ① **Front baffle x1**
18 mm Baltic birch, CNC: cut-outs and rebates
- ② **Back x1**
18 mm birch: port hole, amplifier cut-out, rebate
- ③ **Sides x2**
18 mm birch
- ④ **Top panel x1**
18 mm birch: the 14 mm cable hole, 4 dowel holes
- ⑤ **Bottom panel x1**
18 mm birch
- ⑥ **Window brace x1**
18 mm birch
- ⑦ **Mid shelf and divider x2**
18 mm birch; the divider's 12 mm cable hole
- ⑧ **Gable block x1**
laminated birch, CNC, or printed in six pieces (step 1)
- ⑨ **Waveguide insert x1**
SLA tough resin or MJF nylon; magnets, pins
- ⑩ **Port x1**
printed tube and flare collar, 92 bore
- ⑪ **Amplifier box x3**
birch floor, lid (3 glands) and front; sealed
- ⑫ **Woofer x1**
Dayton Audio RSS315HF-4, 12 in, 4 ohm
- ⑬ **Trim rings x2**
printed, satin black, over each frame, screws
- ⑭ **Midrange x1**
SB Acoustics Satori MR16P-8, 6.5 in
- ⑮ **Tweeter x1**
SB Acoustics Satori TW29DN-B, faceplate off
- ⑯ **Amplifier x1**
Hypex FusionAmp FA253, DSP, 3 channels
- ⑰ **Tweeter retainer x1**
printed sleeve over the motor, 3 screws

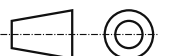
Glue-up order: G1 to G7, sheet 8

Fixings, a speaker (bom.csv)

M4 T-nuts and M4 x 20 low button heads (F1)	12
M16 x 1.5 cable glands and their lock nuts	3
magnets ø12 x 4 (insert and pocket)	8
pins ø6 x 10	2
fluted dowels 10 x 28 (F3)	4
Molex Mini-Fit Jr. 2-circuit plug and socket (3.1)	1 pair
3.0 x 12 thread-forming screws for plastics (sleeve)	3
steel dowel pins ø3 x 16 (the insert's halves)	2
foam gaskets 1.5 and 0.5 mm, EPDM tape 3 mm (the plate)	a strip each
4.3 x 16 self-tapping pan heads (the plate)	as drilled

Exploded view and parts

earmilk floorstander. Isometric, not to scale: the parts pulled apart along the way they go together, and what each one is.

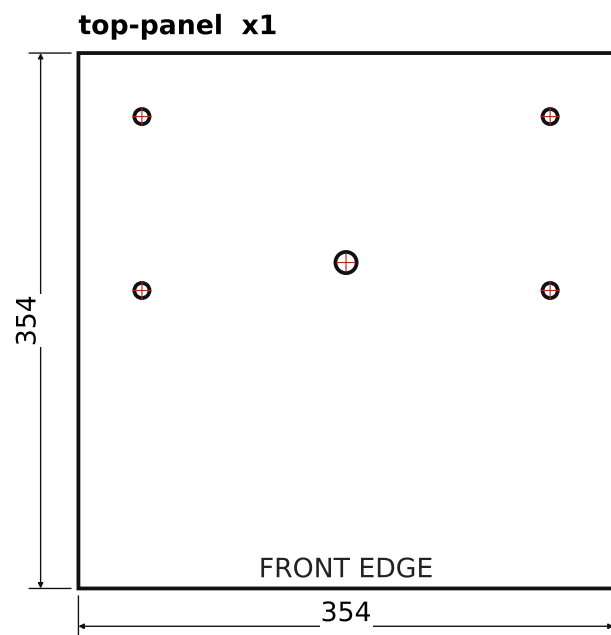


mm | issue 2026-10-09 | from fab/sheets.py (the CAD)

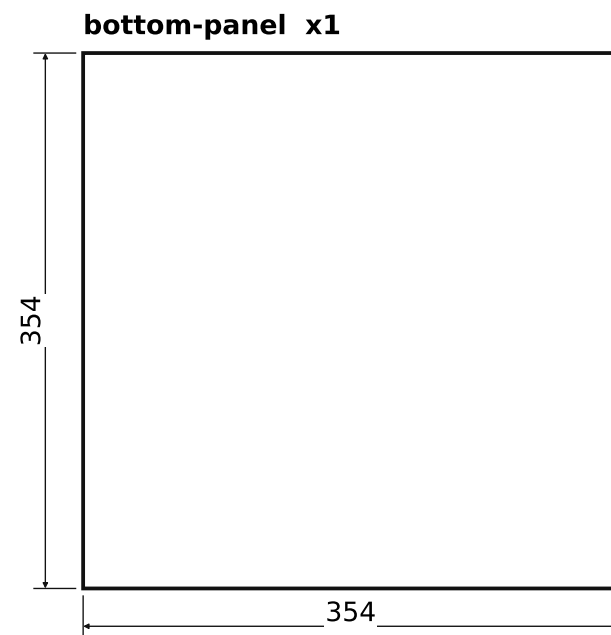
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

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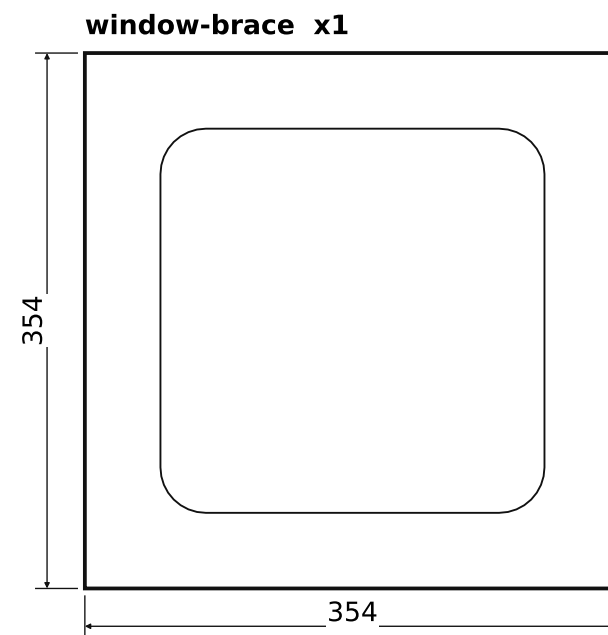
EM-FS-005 rev D



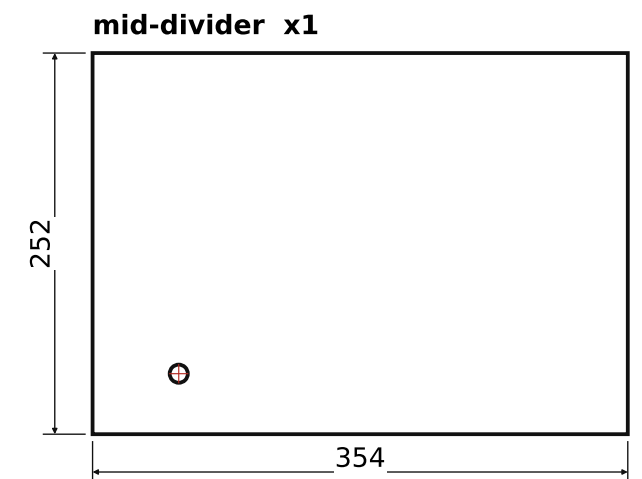
ø14 at (177, 215.5): cut inside
4 x ø10 at (42, 197), (312, 197), (42, 312),
(312, 312): drill ø10, 10 deep
upper face up, FRONT EDGE to the open front
(the ø14 hole 215.5 from it and 138.5 from the
back): turned round, the channel opens under
the insert and the dowels miss the gable;
dowels register the gable block; the tweeter
cable's hole is sealed with silicone from the
bay after wiring



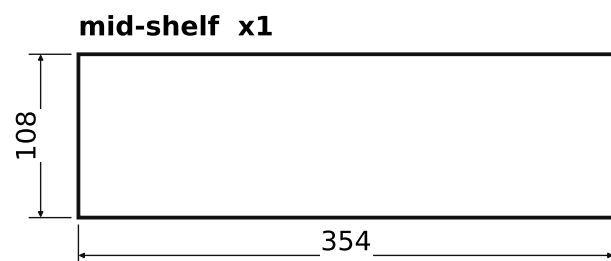
either face



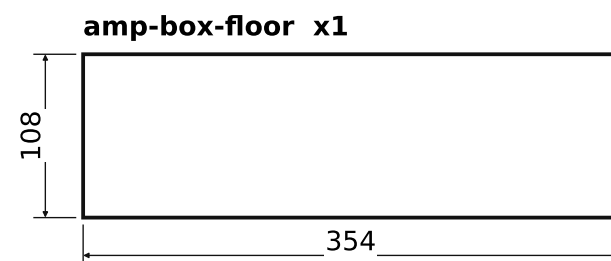
cut inside: 50 to 304 across, 50 to 304 up, R30
corners
sits at z 500 to 518, glued to all four walls



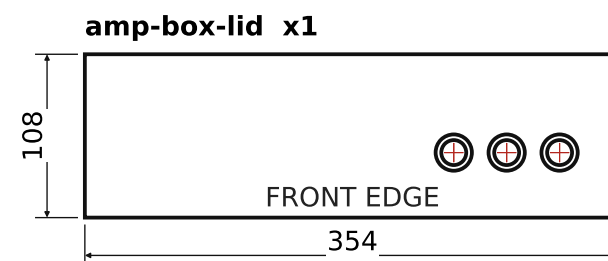
ø12 at (57, 40): cut inside
y 108 to 126, z 590 to 842; seal the wire hole
after wiring



z 572 to 590, from the baffle back to under the
divider

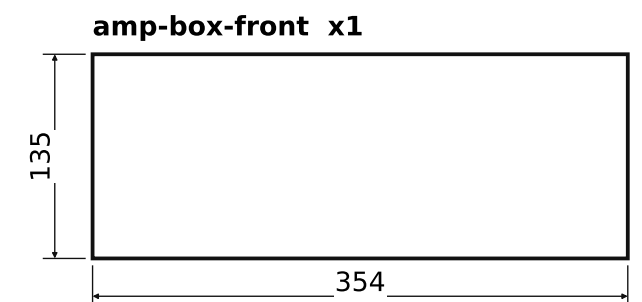


the amplifier box's floor, z 99.5 to 117.5,
against the back



3 x ø16.5 at (314, 43), (279, 43), (244, 43):
cut inside
3 x ø24 at (314, 43), (279, 43), (244, 43):
pocket 8 mm deep from the underside, concentric
with each gland's hole (cut the holes, turn the
panel over, centre on them; before the lid goes
in)

the amplifier box's lid, z 252.5 to 270.5, its
front edge (the bottom of the drawing) over the
box's front; 3 ø16.5 holes for M16 x 1.5
glands, one per cable, 314, 279, 244 from the
left, 43 from the front edge, each nut (20 AF
or less) epoxied into a ø24 x 8 counterbore
from the underside before the lid goes in



the amplifier box's front, between floor and
lid, 90 in front of the back's inner face; glue
and seal all round

Trim rings, section at 2:1 (stl/trim-ring-*.stl): flat, flush with the finish

woofer: ø314 / ø290 x 3



channel open to the bore to ø307.0, 2 deep

mid: ø165 / ø141 x 3

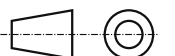


channel from ø145.0 to ø163.0, 2 deep

the channel clears the M4 x 20 heads (F1): 8 across, 1.6 high or less

Panel details

earmilk floorstander. At 1:5: each panel seen from its outer (upper) face, from its
lower-left corner as its DXF; holes from that corner. 18 mm birch.



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Tolerances: panels ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

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EM-FS-006 rev D

Measure first

- M1** Ply: 18 thick; the sides and inner panels are 390 - 2 x 18 = 354 wide. For a real thickness t: 390 - 2t (heights, pockets and rebates unchanged); set WALL = t in params.py and run fab/build.sh again. The gable's layers are t too: 11 for its 195 (one more under t 17.73). The birch under each rebate is t less its depth (woofer t - 9, mid t - 11.5): buy T-nuts whose barrels are 0.5 shorter than that (a CAD check).
- M2** Woofer (Dayton RSS315HF-4): flange 5 thick sets the rebate, 9 deep; the frame's diameter + 1.6 sets the rebate's ø315.6; the cut-out ø282; 8 holes on ø298; the surround at its glue line + 2 sets the trim ring's bore, ø290.
- M3** Mid (SB Satori MR16P-8): flange 7.5 thick sets the rebate, 11.5 deep; the frame's diameter + 1.6 sets the rebate's ø166.6; the cut-out ø140.3; 4 holes on ø154; the surround at its glue line + 2 sets the trim ring's bore, ø141.
- M4** SB Acoustics Satori TW29DN-B: the front ring ø73 sets the bore, ø73.4; its thickness t (6) sets the sleeve's tube, 39.75 - t (33.75) long (0.25 over its gap, so the screws press the gasket), flange 3; the motor ø66 sets the sleeve's bore, ø67; the dome and surround, 45 across, set the throat. Print nothing until it is measured.
- M5** Hypex FusionAmp FA253: the plate 360 x 135 sets the rebate, 361 x 136; its corner radius r sets the rebate's corners, R = r + 0.5 (R4.5 for r 4: a square corner, or one under R2.8, will not seat); its thickness t (3) sets the rebate's depth, t + 1.5 (4.5, over the EPDM pressed), and the screws' length, t + 13 (16); the module sets the cut-out, 336 x 111; the screw holes come from the plate in hand.

Sheet 1: Notes

- 1.1** Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 6 mm, the fin's edges 3 mm. 3 x 3 mm shadow lines at z 110 and z 857.
- 1.2** The drivers sit flush: each rebate is the measured flange + 1.0 (a 1.5 mm closed-cell foam gasket, pressed) + 3 (the ring): woofer 9 for a 5 flange, mid 11.5 for a 7.5 flange; measure each flange before cutting (M2, M3): the rebate is the flange + 4.0 only while the T-nut's barrel + 0.5 of birch stays under it: 9 under the woofer's (barrel 8 or less), 6.5 under the mid's (barrel 6 or less). The screws and T-nuts as F1, pressed in before the glue-up. A printed trim ring over each flange (stl/trim-ring-*.stl, woofer ø314 / ø290 x 3, mid ø165 / ø141 x 3, flat, a channel in its back over the screws' heads: the sections at 2:1 on sheet 6) held by three 5 mm dots of neutral-cure silicone, 0.8 reveal all round. Its inner ø is the surround at its glue line + 2: measure the drivers and print the rings last (the woofer's ring, ø314, needs a bed of 320 or more, or a print service).
- 1.3** The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 935, 176 behind the front face. The shape is set by simulation (fab/out/acoustics/waveguide).
- 1.4** The back (port, amplifier, label) is drawn on sheet 4 with the wiring.
- 1.5** Finish: colour coat and 2K clear; the Facts printed between them.
- 1.6** The cast letters (out/metal/wordmark-letters.*), 1.5 thick and proud: 172.2 x 32.5 of ink, centred across the plinth's front, the badge's centre z 55; on the back the same at z 813. Place and drill them from the 1:1 templates, out/metal/wordmark-template-front.pdf and -back.pdf, after the clear coat.

Sheet 2: The roof

- 2.1** The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round: the pocket +0.2/0, the insert 0/-0.15, so 0.3 to 0.65 a side. Paint only the insert's face: mask its sides, base and back and the pocket's walls. It slides out forward, level, with the tweeter on it.
- 2.2** The tweeter is rear-mounted: it goes in from behind through a ø73.4 bore and its front ring seats on the ring round the throat, its dome and surround filling the 45 mm throat, so the wall runs on from the surround with no step. A printed sleeve over its motor holds it there, screwed to the boss's back face (sheet 3).
- 2.3** Its lead leaves the boss's open back into the connector bay behind it (ø36 x 35 deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed round the cable with neutral-cure silicone from the bay.
- 2.4** Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).
- 2.5** The port: a 92 bore, its tube 100 outside in a ø100.5 hole through the back, centre z 405; its ø112 x 5 flange glued to the finish. 153.3 from the flange's face, with the 18 flare collar 171.3 long. Printed 163.3 and trimmed at its inner end until the impedance dip sits at the tuning: a DATS V3 (or a sound-card impedance jig) clipped to the woofer's terminals, its cable not yet connected; trim 2 mm at a time until the minimum between the two peaks is at 32 Hz (README, step 8).

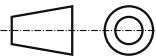
Sheet 3: Fitting the tweeter and the insert

- 3.1** Solder the tweeter's own lead (310 mm of 2 x 0.75 mm2 silicone-insulated flex, twisted (about 2.2 mm each)) to its tabs and crimp the connector's plug on its end: Molex Mini-Fit Jr., 2 circuits, plug housing 39-01-2021 with male terminals 39-00-0041 (18 to 24 AWG) on the tweeter's 0.75 mm2 flex, pin 1 red +. The socket on the cabinet's lead: receptacle housing 39-01-2020 with female terminals 39-00-0077 (16 AWG) on the cabinet's 1.0 mm2 lead. Mated, 24 long or less.
- 3.2** Pass the tweeter in from behind through the ø73.4 bore, dome first, and seat its front ring on a 0.5 mm foam gasket on the throat's seat. Slide the printed retaining sleeve (stl/tweeter-retainer.stl: tube ø73 / ø67 x 33.75 long, flange ø84 x 3) over its motor, lead through it: its flange stands 0.25 off the boss. Drive its 3 3.0 x 12 thread-forming screws for plastics (WN 1411 / PT K30, pan head) into the boss until the flange seats, pressing the gasket by that much: no thread to SB's own screw holes is needed. Measure the motor first: the sleeve's bore is its diameter + 1.
- 3.3** Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Bond the two pins into the insert with epoxy.

- 3.4** The cabinet's lead is fed down the channel from the bay, through the empty pocket, and caught through the woofer's cut-out (it cannot be pushed up from inside). Its socket stands on the bay's floor, the lead's end 77 mm above the top panel's underside (10 past the floor); seal the channel round it with 20 mm of neutral-cure silicone from the bay before the first fitting. To fit: hold the insert just clear of its pocket, reach in and plug the tweeter's lead into the socket (its 310 mm reaches with a hand in the pocket), then slide the insert home, feeding the spare into the bay as a loop, clear of the 1.1 mm between the sleeve's screw heads and the bore's floor.
- 3.5** Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even. Fit: the pocket +0.2/0, the insert 0/-0.15 (0.3 to 0.65 a side); paint only the insert's face.
- 3.6** Service: pull it out by a ribbon loop glued at the back of the 12 x 1.5 x 30 groove under its front edge, unplug, and the tweeter comes out with it. Between times the loop folds back into its groove: only its end shows, under the insert's front edge.
- 3.7** The pocket (gable-block.step): the insert's outline + 0.3 a side (283.9 wide at its base), from the slope to its back wall at y 200.3; the boss's bore ø86.6 to y 222.5; the connector bay ø36 from y 216 to 251 on an axis at z 927; the channel ø14 at x 195, y 233.5 down to the top panel; internal corners R3 or less (the insert's match); dowel holes ø10 x 20 in the base at (60, 215), (330, 215), (60, 330), (330, 330).
- 3.8** Fixings, from the centreline and the insert's base (mirrored in the pocket's wall): magnets ±75.9 across, 30.9 up; ±61.9 across, 75.0 up; pins ±84.1 across, 55.8 up. Magnets ø12 x 4 in ø12.2 x 4.3 holes, epoxied, poles marked; pins ø6 x 10 bonded with epoxy in ø6.1 x 5.5 in the insert, a slip fit in ø6.2 x 5.5 in the wall (their own tolerances, not the title block's: the insert's pin holes ø6.1 +0.05/0, the wall's ø6.2 +0.1/0, the magnets' ø12.2 +0.1/0 x 4.3 +0.2/0, each depth from its own face).
- 3.9** Printing the insert: one piece from a service (SLA tough resin or MJF nylon, a build of 290 x 220 x 130 or more) is the default, with no seam in the waveguide. On a desktop resin printer (218 x 123 x 220) print it in halves split at the centre plane, the section B-B: each half fits only tilted (about 209 x 117 x 213). Join them with two ø3 x 16 steel dowel pins across the split, at y 175.0, z 886.3 and y 190.0, z 977.7, epoxied in ø3.2 holes 8 deep each side; fill and sand the seam on the waveguide's wall flush. MJF prints solid; SLA solid, or hollow to 4 mm walls with two ø3 drain holes in the base face between the pins, plugged with resin: none in the waveguide, the seat or the bore, and none within 5 mm of a magnet, pin, pilot or insert hole.

Notes: measure first; sheets 1 to 3

earmilk floorstander. What to measure before cutting or printing, and the notes sheets 1 to 3 point to by number.



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Tolerances: panels ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

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EM-FS-007 rev D

Sheet 4: Runs, connectors and seals

- 4.1 The amplifier's plate (360 x 135 x 3) lies on 3 mm EPDM tape in its 4.5 deep rebate, R4.5 corners, its module through the 336 x 111 cut-out (R3 corners: a 6 mm cutter or smaller). Drill its screw holes ø3.5 through the 13.5 left under the rebate, from the plate in hand, centred in its flange, 8 or 10 to suit it: 4.3 x 16 self-tapping pan heads (Hypex's 4.3 x 25 kit also works, its tips 7 mm into the sealed box).
- 4.2 Woofer (CH1): from its gland in the amplifier box's lid straight to the woofer's terminals, 0.6 m of 2.5 mm2. Fit 6.3 mm push-on terminals to suit the driver.
- 4.3 Mid (CH2): from its gland up through the brace's window, 30 mm in from its back edge, then through the 12 mm hole in the mid chamber's divider (x 75, z 630), 1.0 m of 1.5 mm2. Seal the hole round the cable with putty: the mid's chamber must stay closed.
- 4.4 Tweeter (CH3): from its gland up through the brace's window to the top panel, through the 14 mm channel in the top panel and the gable block, 1.2 m of 1.0 mm2, ending 77 above the top panel's underside in the socket of the connector (3.1, sheet 3) standing on the bay's floor, fed down from the bay (3.4).
- 4.5 Seal the tweeter cable in its channel with 20 mm of neutral-cure silicone from the bay (sheet 3). The insert's pocket is open to the room through its 0.3 mm seam, so an unsealed channel would be a leak in the woofer's box.
- 4.6 Round-sheathed cable (H05VV-F 2 x 2.5, 2 x 1.5 and 2 x 1.0 mm2, about 9.0, 8.0 and 6.4 across), so each gland grips its cable and the silicone seals round the tweeter's. Inside the amplifier's box cut each of Hypex's harness pairs to about 150 mm and butt-splice it to its round cable (woofer 2.5, mid 1.5, tweeter 1.0 mm2; crimped splices under heat-shrink): only round cable passes the glands. Tie each run with adhesive cable-tie mounts every 150 mm: to the amplifier box's lid beside its gland, to the brace's window edge 30 mm in from its back edge, then over the brace along the top panel's underside (the tweeter's) or the divider's back face (the mid's), 20 mm clear of the port's flare (ø128 round z 405). The brace is glued to the back panel, so no run can follow it.
- 4.7 The amplifier box's lid: 3 M16 x 1.5 long-thread (15 mm) cable glands, one per cable (clamping about 4.5 to 10), in ø16.5 holes 314, 279, 244 from the left side's inner face and 43 from the lid's front edge. Their lock nuts, 20 AF or less (23 across corners) and 5 thick, are epoxied into the ø24 x 8 counterbores under the lid before it goes in (G2), each centred on a gland's body screwed through it and taken out again: a spanner cannot reach a nut sunk 3 deep. After the finish screw the bodies into them from above, feed the cables, then tighten the domes. The box itself is glued and sealed; its front is the woofer chamber's wall.
- 4.8 Polarity, delay and the crossover (about 2.8 kHz tweeter to mid, from the waveguide study) are set in the DSP: wire red to + throughout and let the filters do the rest.

- 4.9 The stencils (out/marks/stencil-*.svg), sprayed through vinyl masks before the clear: SHAKE WELL, 26 mm type, centred on the plinth's back at z 55; OPEN OTHER SIDE, 27 mm type with its arrow, centred on the fin's back face.

Glue-up, in order

- G1 Cut the bottom panel and every inner panel at one fence setting, 354 ±0.2 wide (tighter than the title block's ±0.5: they set and fill the sides' spacing). Lay the back face down. Glue both sides and the bottom to it; clamp across the sides and check the diagonals.
- G2 The amplifier box: epoxy the glands' lock nuts into the lid's counterbores first (sheet 4); then its floor, front and lid between the sides, against the back, the joints sealed; clamp across the sides.
- G3 The window brace at z 500, glued to the sides and the back; clamp across the sides.
- G4 Mark the mid chamber on the sides' inner faces: the shelf's underside 554 above the bottom panel, the divider's front face 90 behind the sides' front edges. Glue the shelf, then the divider, to the marks; clamp across the sides. Check the divider's top with a straightedge across the sides at z 842 and plane it 0.3 below, so the top panel slides over it.
- G5 Line the mid chamber with 10 mm wool or polyester felt (spray adhesive): the divider's front face, the shelf's top, the sides between them, and the top panel's underside over the chamber before it goes in. Leave the front's back face bare.
- G6 The top panel, dowel holes up and its FRONT EDGE to the open front (sheet 6), slid in from the front and glued to the sides and the back; just before, lay a 5 mm bead of PU sealant along the divider's top edge, the mid's chamber's fourth wall (check it after G7 through the mid's cut-out); clamp across the sides.
- G7 Plane any inner panel's front edge that stands proud of the sides flush. Press the T-nuts into the front baffle's inside face; glue the baffle on last, onto the sides, the bottom, the top panel and every inner panel; clamp front to back. When the glue has set, sand the front's top edge flush with the top panel (F2), then fix the gable block (F3).

Fixings and the pocket's floor

- F1 Drivers: M4 x 20 low-profile (wafer) button head, A2, the head 1.6 high or less and 8 across or less (ISO 7380 is 2.2 high: the trim ring's channel is 2 deep). M4 T-nuts, flange 15 or less across (woofer), 12 or less across (mid), barrel 8 or less (woofer), 6 or less (mid), pressed in before the glue-up.

- F2 Before the gable block goes on, sand the front panel's top edge flush with the top panel to 0.1: the insert sits on both. Mask the pocket's floor with the insert's sides, base and back when you spray: paint only the insert's face.
- F3 The gable block on four 10 x 28 fluted dowels (10 deep in the top panel, 20 in the block): a birch block with wood glue; a printed one with epoxy (3M DP420) on its base, scuffed with 120 grit, and on the dowels: wood glue does not hold cured resin.

Notes: sheets 4 to 6; glue-up and fixings

earmilk floorstander. The notes sheets 4 to 6 point to by number; the order of the glue-up; the fixings.

